

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, V: THE DEGENERACY STRUCTURE OF THE IWAHORI TOWER AND THE EULER CORRECTION

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ABSTRACT. The congruence-model (Iwahori path) tower $Y_0 \leftarrow Y_1 \leftarrow \cdots$ of [II, §5] is non-normal, carries no deck action, and obeys the exact pseudo-Iwasawa law $\nu_p = \mu p^k - k + \nu_0$ with $\lambda = -1$, impossible over any commutative $\mathbb{Z}_p[[T]]$ base ([II, Rem. 5.2], [III, §2]). This paper equips that tower with its canonical algebraic structure and identifies, without fabricating a theory that does not yet exist, exactly what any future non-commutative characteristic formalism must specialize to.

The two degeneracy maps — initial and terminal truncation of non-backtracking paths, the graph avatars of $X_0(p^{k+1}) \rightrightarrows X_0(p^k)$ — are packaged into an incidence calculus with six exact identities, among them $\tau_* \eta^* = A_k$ (the level- k Hecke operator realized through level $k+1$; at the bottom, $\tau_* \eta^* = T_p$ of [I]) and $\tau_* \tau^* = \eta_* \eta^* = qI$. Theorem 2.1 computes the degeneracy complex: $\ker(\tau^* - \eta^*) = \mathbb{Z}1$ and $\operatorname{coker}(\tau_* - \eta_*) \cong \mathbb{Z}$ torsion-free, so the Euler rank $\chi = 1$ at every level — the $-\chi$ of [III, Thm. 2.4] is the H^0 of this complex. Theorem 3.1 shows the tail degeneracy alone intertwines the sandpile Laplacians and the Hecke operators, is surjective on p -primary torsion with kernel of the exact order $q^{n_k(q-1)-1}$ forced by [III, Thm. 2.1], and hence assembles the tower into one pro-module $X_\infty = \varprojlim (\operatorname{coker} \Delta_k \otimes \mathbb{Z}_p, \tau_*)$ carrying a limit Hecke operator; the head degeneracy satisfies the mirror identities for the in-Laplacian tower and the exact asymmetry $\eta_* \Delta_{k+1} = q(\eta_* - \tau_*)$, so the structure is genuinely two-sided. Proposition 3.4 records a second, independent rigorous certificate that no commutative base suffices: $A_k A_k^t \neq A_k^t A_k$ at every computed level. We then state, as axioms rather than as a theorem, the specialization data any characteristic theory over the Iwahori–Hecke base must reproduce, and we decline — for the same reason as in [III] — to define an Akashi-type series whose hypotheses our pro-ring does not satisfy.

STATUS OF THE RESULTS

Theorems 2.1 and 3.1 and Propositions 1.2, 3.2 are proved for a general Eulerian q -regular strongly connected level ≥ 1 (the base level $k = 0$ is the boundary case, treated in Remark 3.3); every identity was additionally machine-verified exactly on the K_4 tower at all levels $k \leq 3$ (Table 1). Proposition 3.4 is a finite exact computation on the fixed tower, hence a proof for it; no general-graph claim is made. Section 4 proves nothing and says so: it is a specification, and Problem 4.1 is open.

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Paper V of the series; Papers I–IV are cited as [I]–[IV]. All computational claims are reproduced by the staged scripts `iw_stage1--3.py` in the ancillary bundle. For the fixed tower over K_4 every “verified” statement below is a finite exact-integer computation and is therefore a proof for that tower; general- Y_0 statements are proved generally where so marked.

1. THE DEGENERACY PAIR AND ITS INCIDENCE CALCULUS

Levels: Y_0 a finite connected $(q+1)$ -regular graph; for $k \geq 1$, V_{Y_k} = the non-backtracking k -paths of Y_0 , with an arc $p \rightarrow p'$ when p' is the shift of a common $(k+1)$ -path; $Y_{k+1} = L(Y_k)$ for $k \geq 1$, each Y_k Eulerian q -regular, $n_k = n(q+1)q^{k-1}$ ([II, §5], [III, §2]).

Definition 1.1 (Degeneracy maps). For $k \geq 1$ let $\tau, \eta: Y_{k+1} \rightarrow Y_k$ be terminal and initial truncation of paths, $\tau(v_0 \cdots v_{k+1}) = (v_0 \cdots v_k)$, $\eta(v_0 \cdots v_{k+1}) = (v_1 \cdots v_{k+1})$; under $V_{Y_{k+1}} = \text{Arcs}(Y_k)$ these are the tail and head maps. Write $\tau_*, \eta_* \in M_{n_k \times n_{k+1}}(\mathbb{Z})$ for the pushforwards (sum over fibres) and $\tau^* = \tau_*^t$, $\eta^* = \eta_*^t$ for the pullbacks; thus $\tau_* = D_{\text{out}}$, $\eta_* = D_{\text{in}}$, the incidence matrices of [III, Thm. 2.1].

Proposition 1.2 (Incidence calculus). *For every $k \geq 1$:*

$$\tau_* \eta^* = A_k, \quad \eta_* \tau^* = A_k^t, \quad \eta^* \tau_* = A_{k+1}, \quad \tau^* \eta_* = A_{k+1}^t, \quad \tau_* \tau^* = \eta_* \eta^* = q I_{n_k}.$$

At the boundary level, with $\tau, \eta: Y_1 \rightarrow Y_0$ the endpoint maps on oriented edges, $\tau_ \eta^* = T_p$, the Hecke operator of [I].*

Proof. Entrywise bookkeeping of tails and heads; the middle two are [III, Thm. 2.1, Step 1], the outer two their transposes, the last two the regularity of degrees. The boundary identity is $ST^t = T_p$ in the notation of [I, Thm. 3.7]. All six verified as exact matrix identities at $k = 1, 2$ (Table 1). $\square$

The pair (τ_*, η_*) is the graph avatar of the modular degeneracy pair $X_0(p^{k+1}) \rightrightarrows X_0(p^k)$, and Proposition 1.2 is the statement that the level- k Hecke operator is the correspondence $\tau_* \circ \eta^*$ computed through level $k+1$ — the tower knows its own Hecke theory.

2. THE DEGENERACY COMPLEX AND THE EULER RANK

Theorem 2.1 (Euler rank of the degeneracy complex). *For every $k \geq 1$, with $\partial_k := \tau_* - \eta_*: \mathbb{Z}^{V_{Y_{k+1}}} \rightarrow \mathbb{Z}^{V_{Y_k}}$ and $\partial_k^* := \tau^* - \eta^*$:*

- (i) $\ker \partial_k^* = \mathbb{Z} \cdot \mathbf{1}$: *the only pulled-back-equal functions are the constants — the Perron–Eisenstein line, of rank $\chi_k = 1$;*
- (ii) $\text{coker } \partial_k \cong \mathbb{Z}$ *via the degree functional, and it is torsion-free: the image of ∂_k is exactly the augmentation-zero sublattice.*

Hence the Euler rank of the two-term degeneracy complex is $\chi = 1$ at every level, and the correction $-\chi \cdot k = -k$ of [III, Thm. 2.4] is the accumulated H^0 of this complex, once per level.

Proof. (i) $(\partial_k^* x)_{(a \rightarrow b)} = x_a - x_b$; vanishing forces x constant along arcs, hence constant by strong connectivity. (ii) $\partial_k(e_{(a \rightarrow b)}) = e_a - e_b$; over a connected digraph the differences $e_a - e_b$ along arcs generate, integrally, the full kernel of the degree map (telescope along paths; any z with $\sum z_i = 0$ is an integer combination of consecutive differences), so $\text{im } \partial_k = \ker(\text{deg})$ and $\text{coker} \cong \mathbb{Z}$ with no torsion. Machine confirmation: the Smith form of ∂_k has exactly $n_k - 1$ unit invariants at $k = 1, 2$ (Table 1). $\square$

3. THE TAIL NORM AND THE PRO-MODULE

Theorem 3.1 (Tail norm). *For every $k \geq 1$, with $\Delta_k = qI - A_k$ the out-Laplacian:*

- (i) $\tau_* \Delta_{k+1} = \Delta_k \tau_*$ *and $\tau_* A_{k+1} = A_k \tau_*$;*
- (ii) τ_* *is surjective and preserves the degree functional; hence the induced map on sandpile cokernels is surjective, restricts to a surjection of p -primary torsion, and its kernel there has order exactly $q^{n_k(q-1)-1}$, the increment proved in [III, Thm. 2.1];*

- (iii) consequently $X_\infty := \varprojlim_k (\text{coker}_{\mathbb{Z}}(\Delta_k) \otimes \mathbb{Z}_p, \tau_*)$ is a well-defined pro-module carrying the limit Hecke operator $A_\infty := \varprojlim A_k$, and its torsion-size sequence is the pseudo-Iwasawa law $\text{ord}_p = \mu p^k - k + \nu_0$, $\mu = n(q+1)/q$, of [III, Thm. 2.2] — now the size function of a single pro-object rather than a sequence of unrelated groups.

Proof. (i) $\tau_* A_{k+1} = D_{\text{out}} D_{\text{in}}^t D_{\text{out}} = A_k D_{\text{out}} = A_k \tau_*$ by associativity and Proposition 1.2; subtract from $q\tau_*$. (ii) Every level- k vertex extends, so τ_* hits every basis vector; $\deg \circ \tau_* = \deg$ since fibres are summed; torsion-surjectivity then follows exactly as in [IV, Lem. 1.3] (the kernel of $\deg$ is the torsion), and the kernel order is forced by counting against [III, Thm. 2.1]. (iii) Inverse limits over surjections of compact modules; the Hecke compatibility is (i). Explicit machine confirmation of (ii) at the transition $2 \rightarrow 1$: surjective on the 2-part with kernel order exactly $2^{11} = 2^{n_1-1}$, the orders 7, 18 matching [II, Rem. 5.2] (Table 1). $\square$

Proposition 3.2 (Head mirror, and the exact asymmetry). *For every $k \geq 1$: η_* intertwines the in-Laplacians, $\eta_*(qI - A_{k+1}^t) = (qI - A_k^t)\eta_*$, so the mirrored construction yields the in-sandpile pro-module X_∞^{in} ; on the out-tower, however,*

$$\eta_* \Delta_{k+1} = q(\eta_* - \tau_*) = -q \partial_k,$$

so η_* does not descend to the out-cokernels: the failure is exactly q times the degeneracy boundary of Theorem 2.1. The tower is thus canonically two-sided — a (τ, η) -bimodule system, not a single inverse system — and the obstruction to collapsing the two sides is again the Euler complex.

Proof. $\eta_* A_{k+1}^t = D_{\text{in}} D_{\text{out}}^t D_{\text{in}} = A_k^t D_{\text{in}}$; and $\eta_* \Delta_{k+1} = qD_{\text{in}} - D_{\text{in}} D_{\text{in}}^t D_{\text{out}} = q(D_{\text{in}} - D_{\text{out}})$ by $D_{\text{in}} D_{\text{in}}^t = qI$. Verified at $k = 1, 2$. $\square$

Remark 3.3 (The boundary level). At $k = 0$ the base is undirected and $(q+1)$ -regular, and the clean intertwining fails by exactly the degeneracy boundary: $\tau_* \Delta_1 - \Delta_0 \tau_* = -(\tau_* - \eta_*) = -\partial_0$, verified as an exact identity. The Eisenstein special layer of the whole series — the $(q+1)$ -versus- q mismatch of [I, Rem. 1.8], the shared Perron root of [II], the excluded trivial eigenvalue of [III] — reappears here as the unique level where the norm formalism acquires a correction term, and the correction is once more ∂ .

Proposition 3.4 (Second non-commutativity certificate). *On the K_4 tower, $A_k A_k^t \neq A_k^t A_k$ for $k = 1, 2, 3$ (exact computation, hence a proof for this tower). Together with the growth-theoretic certificate $\lambda = -1 < 0$ of [III, Prop. 2.3], this gives two independent rigorous witnesses that the level operator algebras $\mathbb{Z}\langle A_k, A_k^t \rangle$, and a fortiori any base ring over which the tower's characteristic theory could live, are non-commutative: the first says the growth cannot come from $\mathbb{Z}_p[[T]]$, the second that the operators themselves do not commute with their adjoints.*

4. WHAT A CHARACTERISTIC THEORY MUST SPECIALIZE TO

We now state — as a specification, not a theorem — the data pinned down by Papers II–V that any characteristic formalism Char for (τ, η) -towers over the (pro-)Iwahori–Hecke base must reproduce:

- (A1) a multiplicative rank-density term recovering $\mu = n(q+1)/q$, the arc proliferation of [III, Thm. 2.2];
- (A2) an additive Euler term $-\chi \cdot k$ with $\chi = \text{rk ker}(\tau^* - \eta^*)$, the H^0 of the degeneracy complex (Theorem 2.1) — intrinsic to the base, external to any commutative characteristic ideal;
- (A3) on sub-towers carrying an abelian deck action, reduction to the commutative characteristic ideal $(\det D(1+T))$ of [IV, Thm. 2.1], with the Newton-polygon evaluation of [II, Rem. 5.4].

Problem 4.1. Construct Char satisfying (A1)–(A3) on the pro-module $(X_\infty, A_\infty, \tau, \eta)$ of Theorem 3.1.

We explicitly decline to define an Akashi-type series here. The non-commutative Iwasawa formalism in which Akashi series live is built over Iwasawa algebras of compact p -adic Lie groups [1]; our pro-ring is the inverse limit of the finite-dimensional, non-normal level algebras of Proposition 3.4, generated by degeneracy correspondences rather than by a group, and none of the hypotheses of that formalism are available. Writing down an “Akashi series” whose only property is to output the already-proved growth law would be circular dressing, and this series does not do that ([III, Rem. 2.6]). What Papers II–V have done instead is to make Problem 4.1 well-posed: the category, the operators, the two invariants any answer must produce, and the commutative shadow it must reduce to, are now all theorems.

5. THE VERIFICATION BATTERY

check	levels	result
(F) six incidence identities, incl. $\tau_*\eta^* = A_k$, $\tau_*\tau^* = \eta_*\eta^* = qI$	$k = 1, 2$	exact
(T) $\tau_*\Delta_{k+1} = \Delta_k\tau_*$, $\tau_*A_{k+1} = A_k\tau_*$	$k = 1, 2$	exact
(H) in-mirror for η_* ; asymmetry $\eta_*\Delta_{k+1} = q(\eta_* - \tau_*)$	$k = 1, 2$	exact
(E) $\ker(\tau^* - \eta^*) = \mathbb{Z}\mathbf{1}$; SNF($\tau_* - \eta_*$): $n_k - 1$ units	$k = 1, 2$	exact
(N) tail norm on 2-part: surjective, $ \ker = 2^{11} = 2^{n_1-1}$	$2 \rightarrow 1$	exact
(C) $A_k A_k^t \neq A_k^t A_k$	$k = 1, 2, 3$	exact
(Z) $\tau_*\Delta_1 - \Delta_0\tau_* = -\partial_0$; $\tau_*\eta^* = T_p$ at the bottom	$0 \rightarrow 1$	exact

TABLE 1. Exact integer verification (scripts `iw_stage1--3.py`) on the K_4 tower, $q = 2$, levels ≤ 3 (up to 48 vertices). Each general lemma is proved in the text; for the fixed tower each line is itself a proof.

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Unrefereed preprint. All numbered statements are proved, with Proposition 3.4 proved for the fixed K_4 tower by exact computation and asserted for no other graph. Section 4 is a specification; Problem 4.1 is open, and the paper’s deliberate refusal to define an Akashi-type invariant is explained there. The single external citation [1] is contextual and load-bearing for nothing; its bibliographic data are now primary-source verified.

REFERENCES

- [1] J. Coates, T. Fukaya, K. Kato, R. Sujatha and O. Venjakob, *The GL_2 main conjecture for elliptic curves without complex multiplication*, Publ. Math. Inst. Hautes Études Sci. **101** (2005), 163–208. [Primary-source verified: the IHÉS journal page, Volume 101 (2005), pp. 163–208, authors and title as listed.]

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